39th Annual

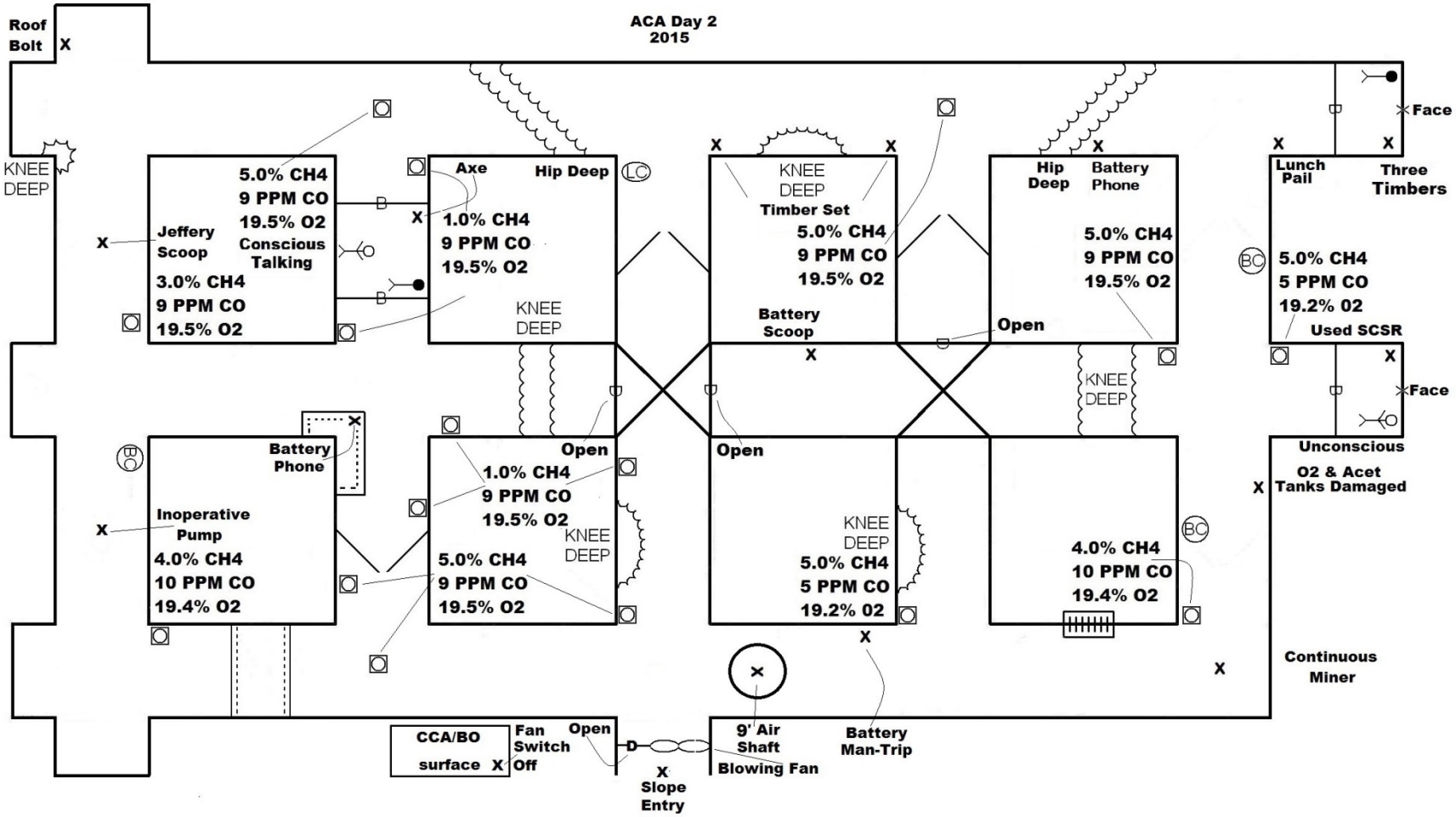
Alabama Coal Association



2015 Day 2 Judges Book

FUTURE MADE VISIBLE

Bevill State Community College


ACA Day 2
2015
Roof Bolt
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Jeffery Scoop
Conscious Talking
3.0% CH4
9 PPM CO
19.5% O2
Axe
Hip Deep
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
KNEE DEEP
Timber Set
5.0% CH4
9 PPM CO
19.5% O2
Battery Scoop
Hip Deep
Battery Phone
5.0% CH4
9 PPM CO
19.5% O2
Open
KNEE DEEP
Face
Lunch Pail
Three Timbers
5.0% CH4
5 PPM CO
19.2% 02
Used SCSR
Face
Unconscious
O2 & Acet Tanks Damaged
Battery Phone
Inoperative Pump
4.0% CH4
10 PPM CO
19.4% O2
Open
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Open
KNEE DEEP
5.0% CH4
5 PPM CO
19.2% 02
4.0% CH4
10 PPM CO
19.4% O2
Continuous Miner
CCA/BO
surface
Fan Switch Off
Open
Slope Entry
9' Air Shaft
Blowing Fan
Battery Man-Trip

2015 ACA

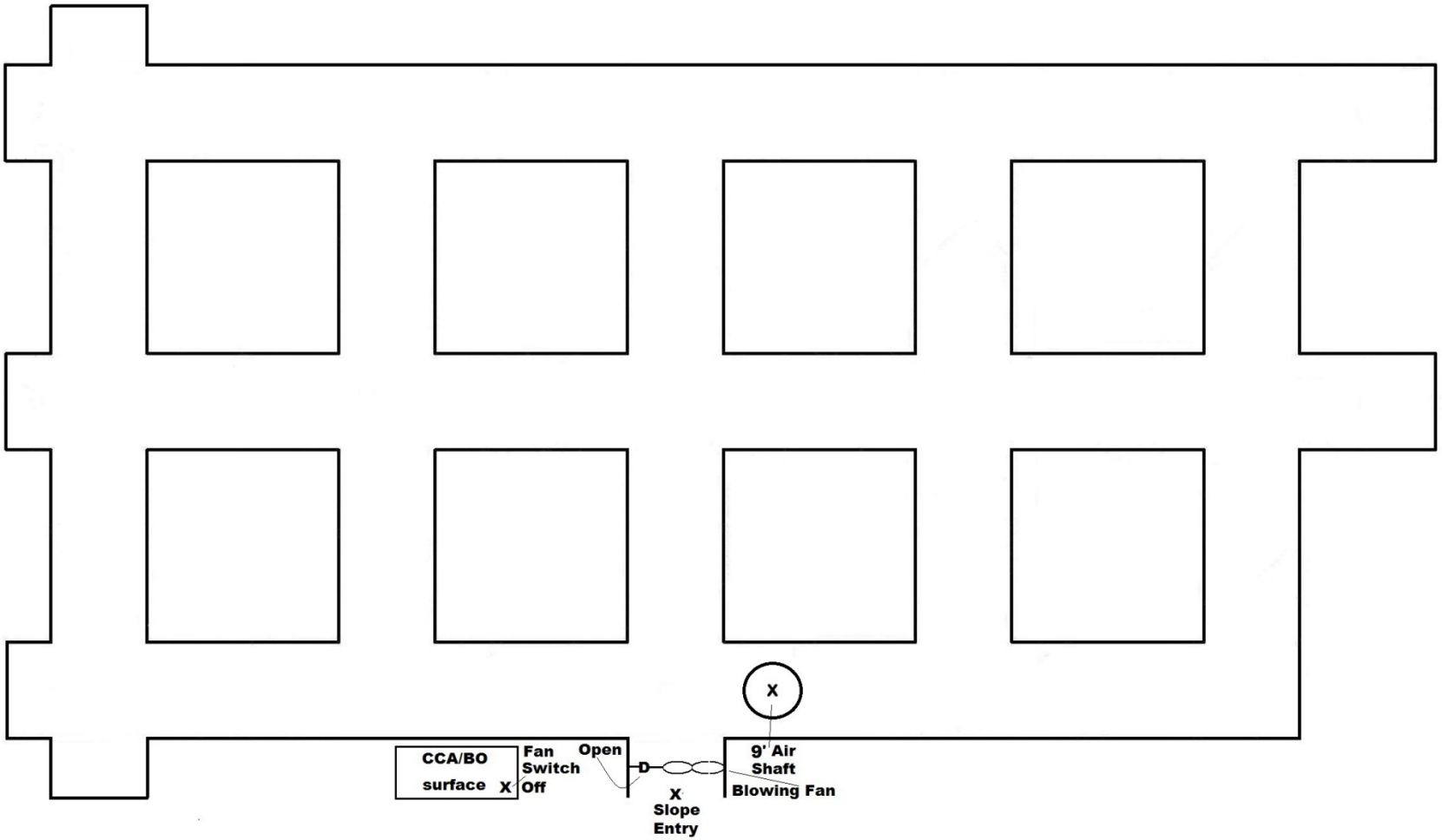
DAY 2 PROBLEM

GENTLEMEN, WE THANK YOU FOR ANSWERING OUR CALL FOR HELP…….

Four people were working on the No.1 unit early this morning when the explosion door on the fan blew open and the fan stopped. The outside dispatcher called underground and could not get anyone by phone or radio. The RP went down the slope/fan entry and found high methane levels inby the first crosscut underground and decided not to go any further inby and came back outside. You are on the surface and can travel the slope entry into the mine. The mine is driving five entries inby the slope at this time and plans to turn to the north and drive three entries in the near future. The mine just completed an airshaft that can be used for ventilation if needed. A fresh air base has been established here on the surface.

All authorities have been notified and are here on site. A backup team is present and ready. The mining maps are up to date and will be given to you when you start the clock.

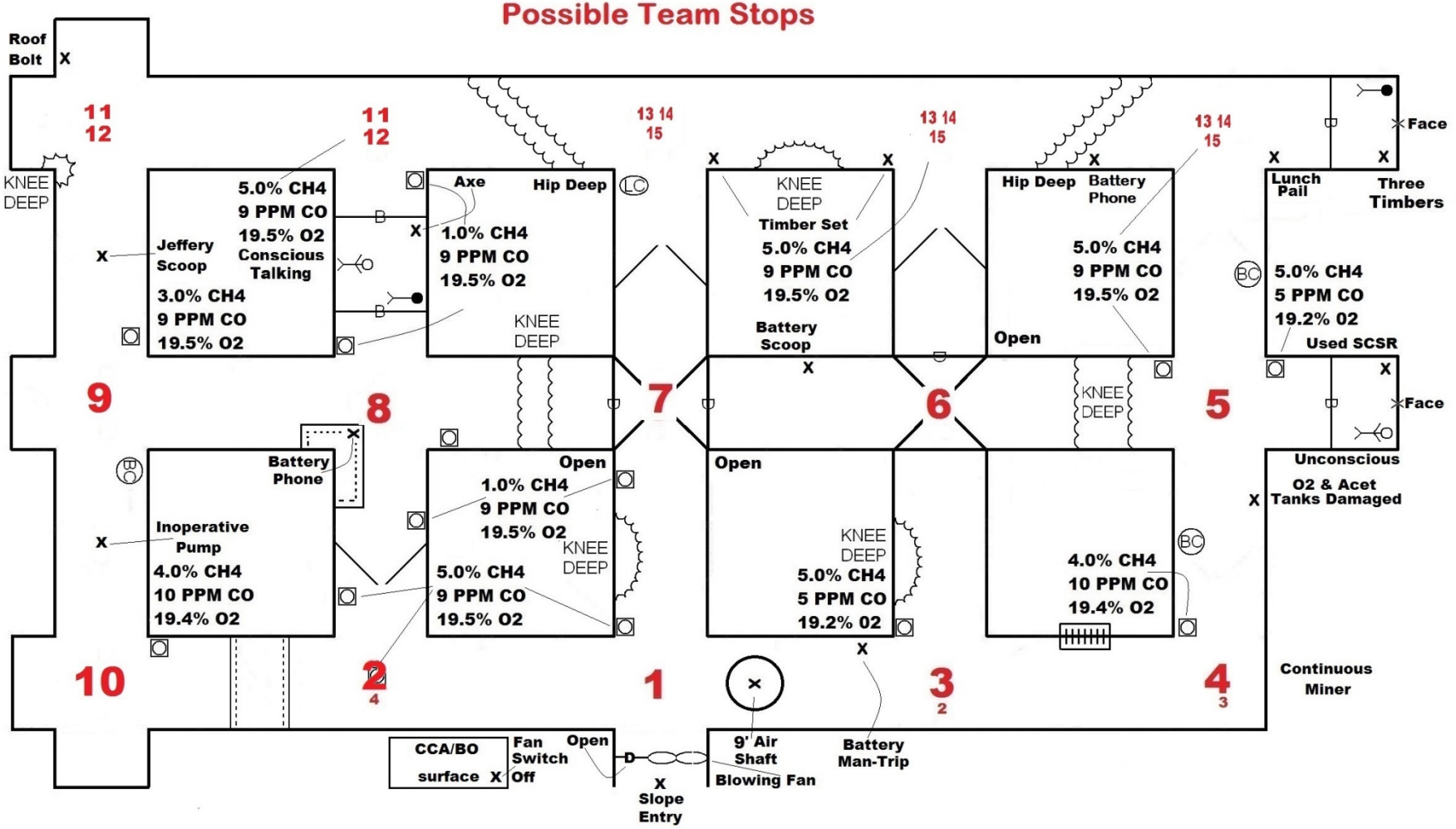
Please be careful during your exploration ……


X
CCA/BO
surface
X
Fan
Switch
Off
Open
D
X
Slope
Entry
9' Air
Shaft
Blowing Fan

WRITTEN INSTRUCTIONS
2015 ACA Day 2

- ACCOUNT FOR ALL MISSING MINERS
- BRING ALL SURVIVORS TO THE SURFACE
- THE BLOWING FAN CAN BE STOPPED AND STARTED BY USING THE FAN SWITCH INSIDE THE CCA/BO STATION ON THE SURFACE
- THE BLOWING FAN CANNOT BE REVERSED AND CANNOT BE STALLED WHILE IN OPERATION
- YOUR TEAM WILL HAVE 80 MIN. TO COMPLETE THE PROBLEM

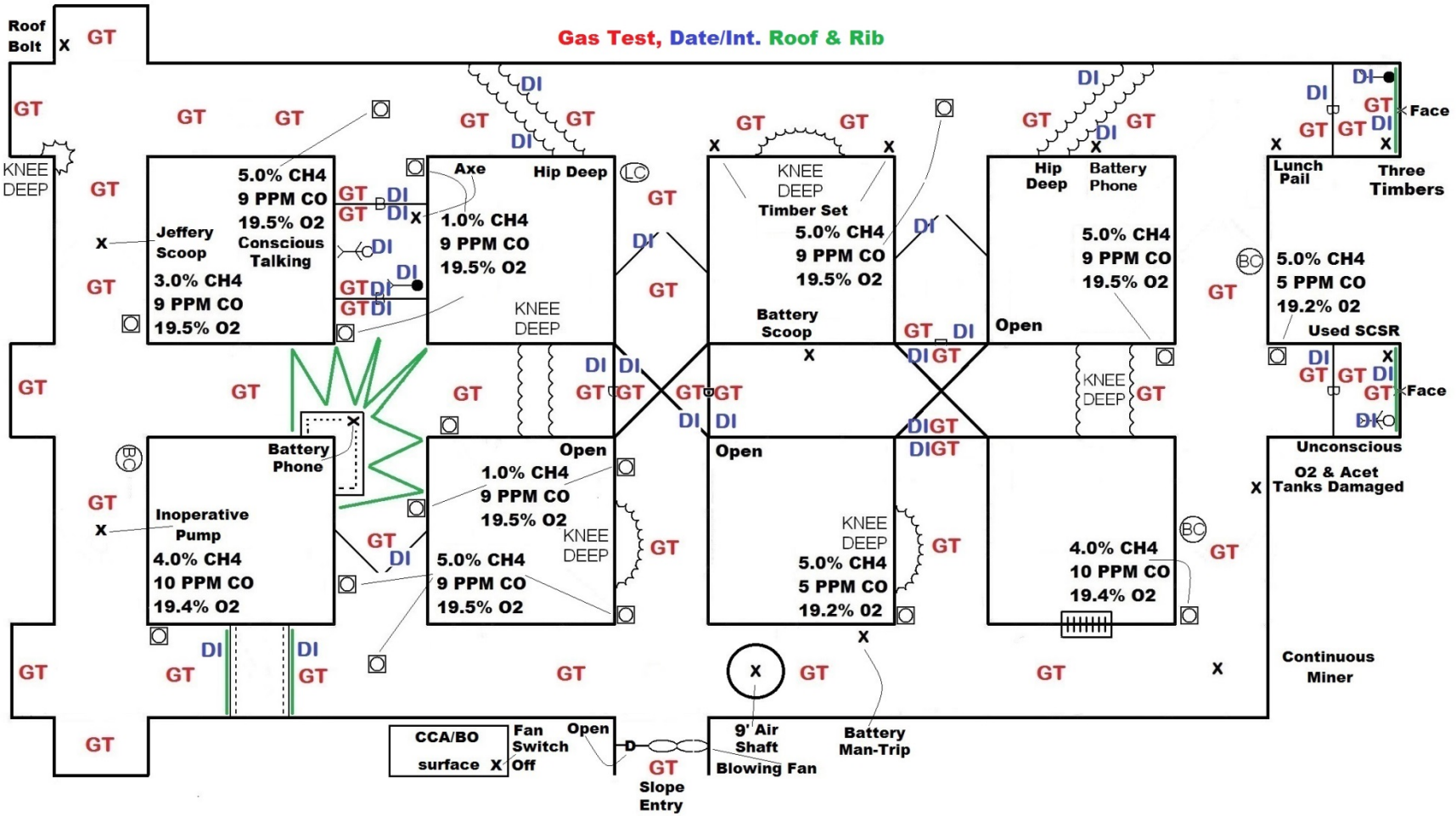
Possible Team Stops

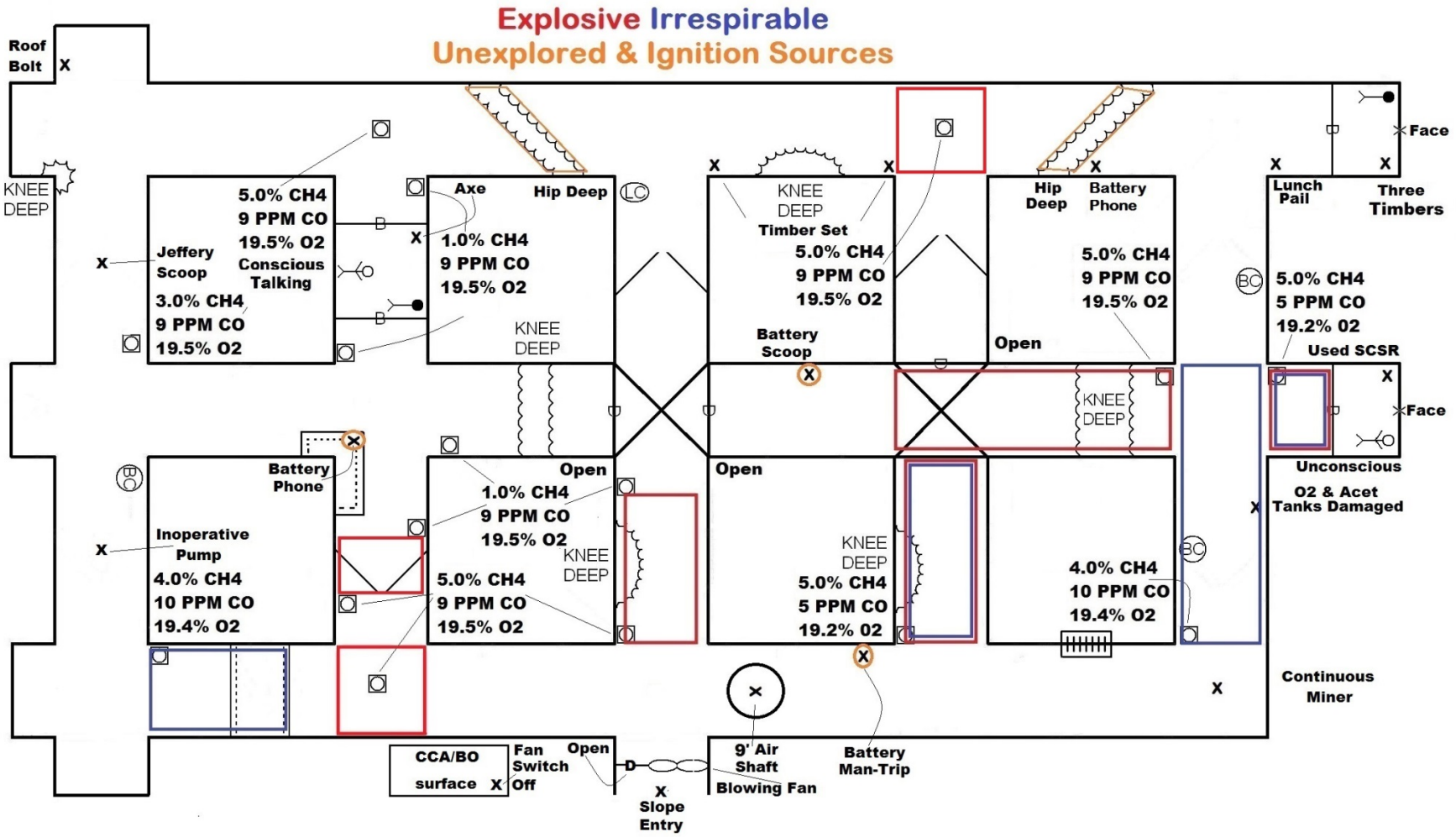


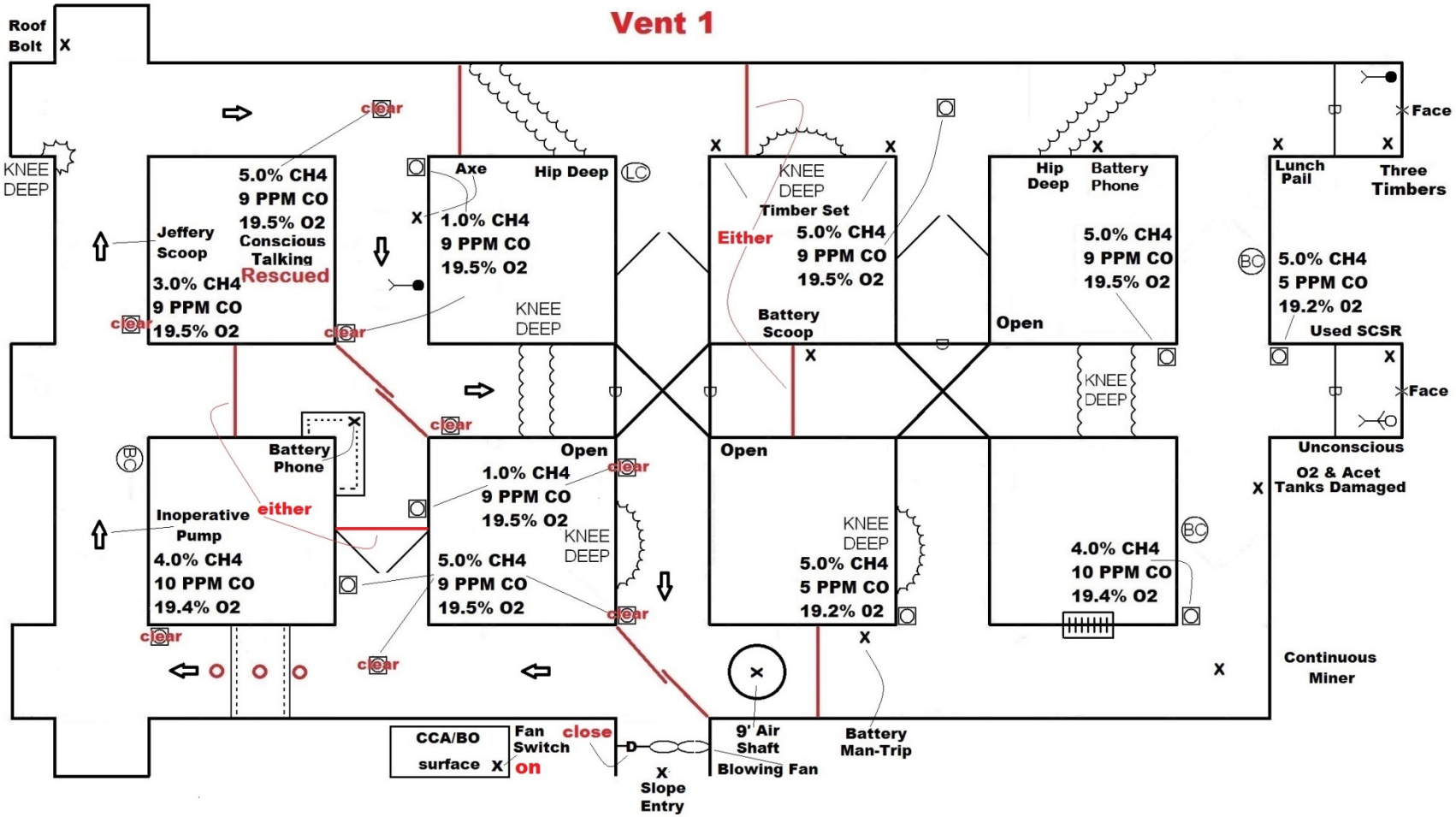
Conscious Miner Statement

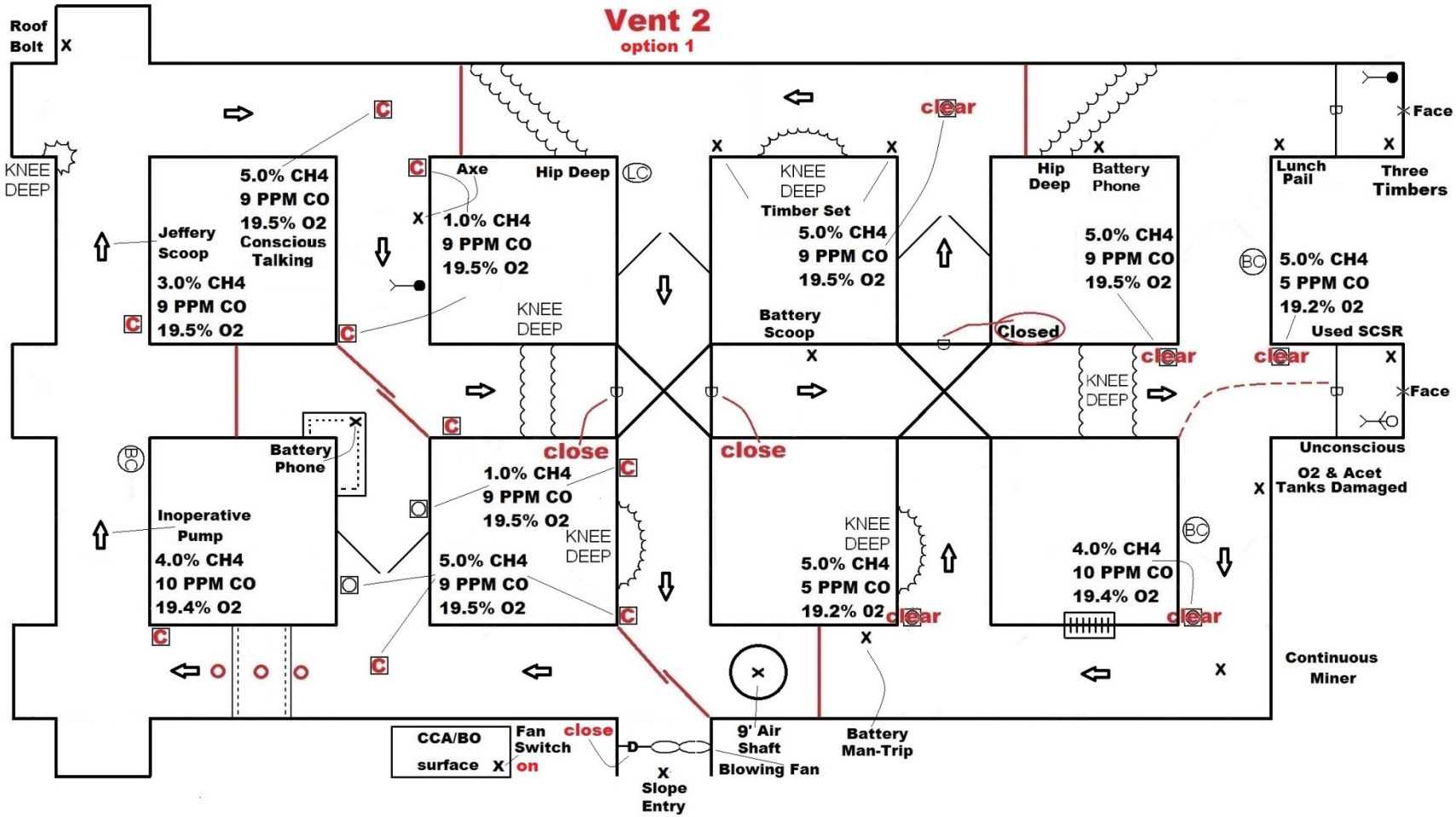
Statement behind both barricades in second crosscut

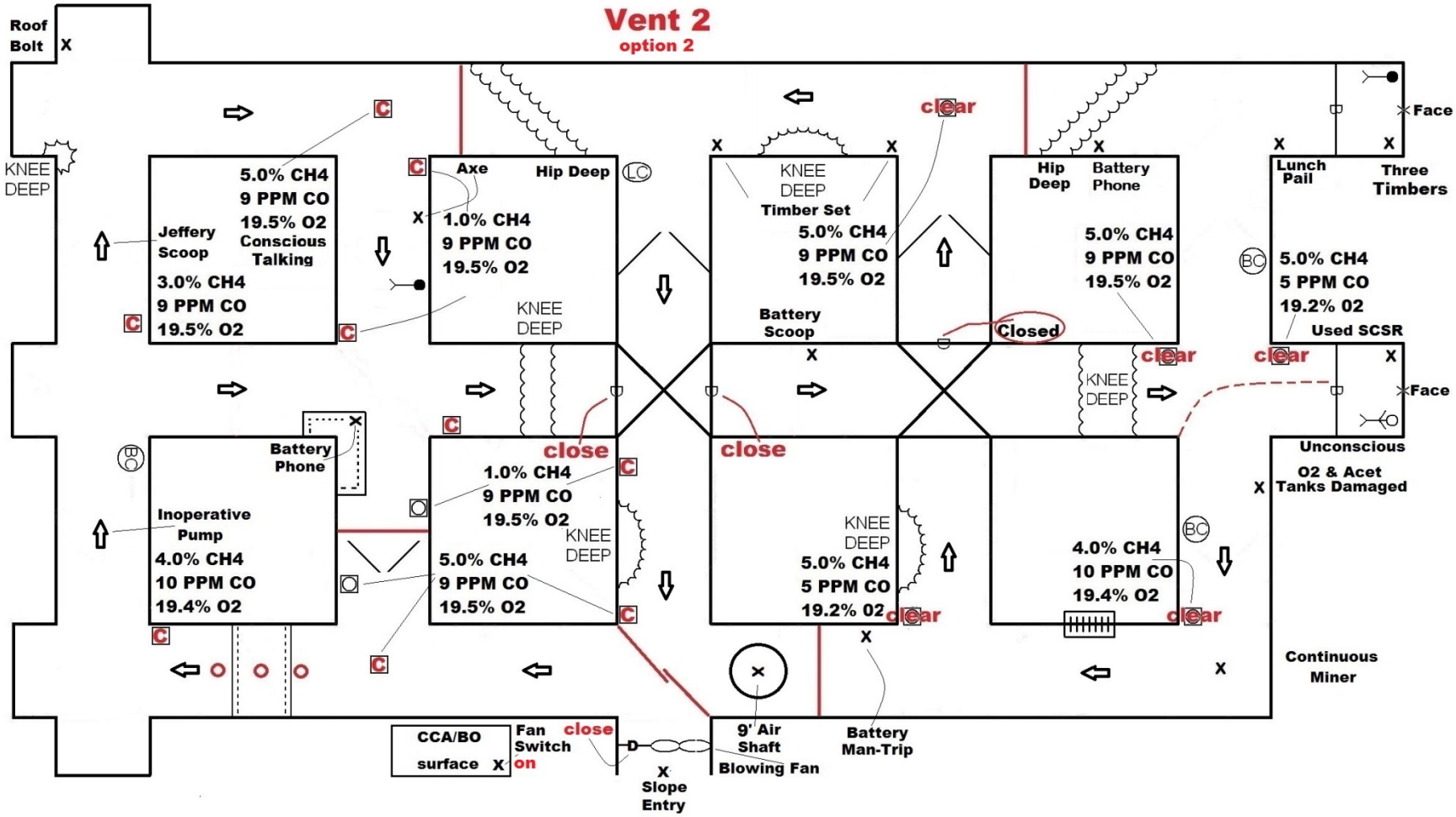
Help! I have a airtight barricade behind me !


Gas Test, Date/Int. Roof & Rib
Roof Bolt
GT
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Conscious Talking
Jeffery Scoop
3.0% CH4
9 PPM CO
19.5% O2
Axe
Hip Deep
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
Timber Set
5.0% CH4
9 PPM CO
19.5% O2
Battery Scoop
Hip Deep
Battery Phone
5.0% CH4
9 PPM CO
19.5% O2
Open
Lunch Pail
Three Timbers
5.0% CH4
5 PPM CO
19.2% 02
Used SCSR
Face
Unconscious
O2 & Acet Tanks Damaged
Battery Phone
Inoperative Pump
4.0% CH4
10 PPM CO
19.4% O2
Open
1.0% CH4
9 PPM CO
19.5% O2
5.0% CH4
9 PPM CO
19.5% O2
Open
KNEE DEEP
5.0% CH4
5 PPM CO
19.2% 02
4.0% CH4
10 PPM CO
19.4% O2
Continuous Miner
CCA/BO
surface
Fan Switch Off
Open
Slope Entry
9' Air Shaft
Blowing Fan
Battery Man-Trip


Explosive Irrespirable
Unexplored & Ignition Sources
Roof Bolt
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Conscious Talking
Jeffery Scoop
3.0% CH4
9 PPM CO
19.5% O2
Axe
Hip Deep
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
KNEE DEEP
Timber Set
5.0% CH4
9 PPM CO
19.5% O2
Battery Scoop
Hip Deep
Battery Phone
5.0% CH4
9 PPM CO
19.5% O2
Open
Face
Lunch Pail
Three Timbers
5.0% CH4
5 PPM CO
19.2% O2
Used SCSR
Face
Unconscious
O2 & Acet Tanks Damaged
KNEE DEEP
Battery Phone
Inoperative Pump
4.0% CH4
10 PPM CO
19.4% O2
Open
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Open
KNEE DEEP
5.0% CH4
5 PPM CO
19.2% O2
4.0% CH4
10 PPM CO
19.4% O2
Continuous Miner
CCA/BO
surface
Fan Switch Off
Open
Slope Entry
9' Air Shaft
Blowing Fan
Battery Man-Trip


Vent 1
Roof Bolt
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Conscious
Talking
Jeffery Scoop
3.0% CH4
9 PPM CO
19.5% O2
Rescued
clear
Axe
Hip Deep
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
Either
KNEE DEEP
Timber Set
5.0% CH4
9 PPM CO
19.5% O2
Battery Scoop
Hip Deep
Battery Phone
5.0% CH4
9 PPM CO
19.5% O2
Open
Lunch Pail
Three Timbers
Face
5.0% CH4
5 PPM CO
19.2% 02
Used SCSR
Unconscious
O2 & Acet Tanks Damaged
Battery Phone
Inoperative Pump
either
4.0% CH4
10 PPM CO
19.4% O2
Open
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Open
KNEE DEEP
5.0% CH4
5 PPM CO
19.2% 02
4.0% CH4
10 PPM CO
19.4% O2
Continuous Miner
CCA/BO
surface
Fan Switch
close
on
Slope Entry
9' Air Shaft
Blowing Fan
Battery Man-Trip


Vent 2
option 1
Roof Bolt
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Conscious Talking
Jeffery Scoop
3.0% CH4
9 PPM CO
19.5% O2
Axe
Hip Deep
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
KNEE DEEP
Timber Set
5.0% CH4
9 PPM CO
19.5% O2
clear
Battery Scoop
Closed
Hip Deep
Battery Phone
5.0% CH4
9 PPM CO
19.5% O2
Lunch Pail
Three Timbers
Face
5.0% CH4
5 PPM CO
19.2% 02
Used SCSR
clear
clear
Face
KNEE DEEP
Unconscious
O2 & Acet Tanks Damaged
Battery Phone
Inoperative Pump
4.0% CH4
10 PPM CO
19.4% O2
close
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
close
KNEE DEEP
5.0% CH4
5 PPM CO
19.2% 02
clear
4.0% CH4
10 PPM CO
19.4% O2
clear
Continuous Miner
CCA/BO
surface
Fan Switch
close
on
Slope Entry
9' Air Shaft
Blowing Fan
Battery Man-Trip


Vent 2
option 2
Roof Bolt
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
Conscious Talking
Jeffery Scoop
3.0% CH4
9 PPM CO
19.5% O2
Axe
Hip Deep
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
KNEE DEEP
Timber Set
5.0% CH4
9 PPM CO
19.5% O2
Battery Scoop
clear
Hip Deep
Battery Phone
5.0% CH4
9 PPM CO
19.5% O2
Closed
Face
Lunch Pail
Three Timbers
5.0% CH4
5 PPM CO
19.2% 02
Used SCSR
clear
clear
Face
KNEE DEEP
Unconscious
O2 & Acet Tanks Damaged
Battery Phone
Inoperative Pump
4.0% CH4
10 PPM CO
19.4% O2
close
close
1.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
5.0% CH4
9 PPM CO
19.5% O2
KNEE DEEP
5.0% CH4
5 PPM CO
19.2% 02
clear
4.0% CH4
10 PPM CO
19.4% O2
clear
Continuous Miner
CCA/BO
surface
Fan Switch
on
close
Slope Entry
9' Air Shaft
Blowing Fan
Battery Man-Trip

Problem Keys

- Team must tie across and behind before traveling inby while exploring ….(five entries in this problem/see team stop map)
- Team must advance in the contaminated entry or adjacent entry while exploring (five entries in this problem/see team stop map)
- No vent required to rescue conscious patient from between the two barricades in second crosscut
- Must find timbers behind barricade in last open before any vent can be done correctly (no vent required for barricade with 3 timbers)
- Team must use 3 timbers at unsafe roof rib to rib (not enough timbers to timber to battery phone/takes 4 timbers for intersection)
- Team must close door at fan. (Fan is inby FAB)
- Team must not move explosive mixture over any battery equipment or unexplored areas